From burst to vessel graph: gating, stabilisation, optical density and Hessian vesselness

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This note sets out the first stages of vessel annotation as equations, following the MICA code where it exists (`ingest_stabilize.py`, `background.py`). Frames that are blurred or show no vessels are gated out (section 1). The burst is split into fixations at saccades and registered within and between fixations, with rotation (section 2). Fixations are clustered by their fitted backgrounds, and each cluster's average is converted to optical density with its own background before the clusters are combined (section 3). Multi-scale Gaussian derivatives then give the Hessian at every pixel and scale (sections 4–5), from which Frangi's vesselness and a width estimate follow (sections 6–7), and vesselness is turned into a graph (section 8). Section 9 shows where this Hessian chain fails, which is why the later stages use oriented cells.

Notation

$I_k(\mathbf{x})$, $k = 1, \dots, N$	the frames of the burst; mono12, so values are DN in $[0, 4095]$
$\mathbf{x} = (x, y)$	pixel position; the image is $W \times H$ pixels
$\mathcal{F}$, $\mathcal{F}^{-1}$	discrete Fourier transform and its inverse
G_σ	2-D Gaussian of standard deviation σ (px); $*$ is convolution
Φ_f	the frames of fixation f; $\mathcal{C}_q$ the fixations of background cluster q
T_k	the transform from stabilised coordinates into frame k
$I'_k(\mathbf{x})$	frame k after stabilisation; $\bar{I}$ an average of stabilised frames
$L = \ln \bar{I}$, B	log image and fitted background, in log units
$\mathrm{OD}(\mathbf{x})$	optical density; $\sigma_n(\mathbf{x})$ its noise

1 Frame gating

Each frame is scored on a copy shrunk by $D = 4$ (block means) and divided by its own mean, so changes of exposure do not move the scores:

$$\tilde{I}_k = \frac{\downarrow_D I_k}{\langle \downarrow_D I_k \rangle_{\mathbf{x}}}. \tag{1}$$

The scores are sharpness, the variance of the Laplacian (Pech-Pacheco et al., 2000), and vessel content, the mean Frangi vesselness (section 6; scales $\sigma \in \{1, 2, 3\}$ px of the shrunk frame, so 4–12 px of the original, dark ridges):

$$s_k = \mathrm{Var}_{\mathbf{x}}\big(\nabla^2 \tilde{I}_k\big), \qquad v_k = \big\langle V(\tilde{I}_k) \big\rangle_{\mathbf{x}}, \tag{2}$$

with ∇^2 the 4-neighbour discrete Laplacian, kernel $\left(\begin{smallmatrix} 0 & -1 & 0 \\ -1 & 4 & -1 \\ 0 & -1 & 0 \end{smallmatrix}\right)$. A frame passes when both scores are at least 0.8 of the burst's median:

$$\mathrm{keep}_k = \big[\, s_k \geq 0.8 \ \mathrm{med}_j \, s_j \,\big] \wedge \big[\, v_k \geq 0.8 \ \mathrm{med}_j \, v_j \,\big]. \tag{3}$$

Thresholds relative to the median adapt to each burst's exposure, noise and content; they assume most frames are good.

Two corrections. As defined, both scores can rank the wrong frames first. Table 1 scores synthetic frames: the same vessels sharp, blurred or absent, and dim frames with 0.3 times the light, whose relative noise after division by the mean is 3.3 times higher.

- **Noise adds to the sharpness.** For white noise of standard deviation $\hat{\sigma}_k$, the Laplacian adds $20\,\hat{\sigma}_k^2$ to s_k (20 is the sum of its squared weights), so a dim, vessel-free frame scored as the sharpest. Subtract the noise term,

$$s'_k = s_k - 20\,\hat{\sigma}_k^2, \qquad \hat{\sigma}_k = \sqrt{\frac{\pi}{2}}\,\frac{1}{6(W-2)(H-2)}\sum_{\mathbf{x}}\left|(\tilde{I}_k * N)(\mathbf{x})\right|, \quad N = \begin{pmatrix} 1 & -2 & 1 \\ -2 & 4 & -2 \\ 1 & -2 & 1 \end{pmatrix}, \tag{4}$$

  with $\hat{\sigma}_k$ from Immerkær's estimator (W, H here the shrunk frame's size).

- **Frangi's c is set per frame.** scikit-image sets c to half the largest S of each frame (at its first scale), so a frame with no vessels is scored against its own noise, and its noise looks vessel-like. Vessel-free frames got the highest mean vesselness. Use one c for the whole burst, from its median frame, or c in noise units (section 6).

frame	s_k (10^{-4})	s'_k (10^{-4})	v_k, own c	v_k, burst c
sharp	8.95	6.77	0.0179	0.0071
blurred, 4 px	4.65	2.64	0.0210	0.0043
blurred, 8 px	2.78	0.74	0.0261	0.0018
no vessels	1.91	−0.01	0.0572	0.0000
no vessels, dim	20.90	−0.14	0.0598	0.0003
sharp, dim	29.30	6.79	0.0179	0.0074

Table 1: Gate scores of synthetic frames, one of each kind (blur in pixels of the full frame; the burst's c taken from the 4 px frame, standing in for the median frame). The raw scores rank a dim, vessel-free frame as the sharpest and vessel-free frames as the most vessel-like. The corrected scores rank sharp, then blurred, then no vessels, whatever the brightness. Check them on real bursts with known blinks before changing the thresholds.

2 Stabilisation

Cleaned copies for registration. Registration runs on cleaned copies of the frames; the frames themselves are only warped at the end. Dividing by the burst's per-pixel mean $\bar{S}(\mathbf{x}) = \langle I_k(\mathbf{x})\rangle_k$ removes what is fixed to the sensor (dust, hot pixels, illumination fall-off), since the moving retina blurs out of the mean. A high-pass then removes slow brightness changes:

$$J_k = \frac{I_k}{\bar{S}} - G_{\sigma_h} * \frac{I_k}{\bar{S}}, \qquad \sigma_h = 20 \text{ px}. \tag{5}$$

Both would otherwise pull the registration towards zero shift. The mean is used, not the median: when most frames sit near one position, the median keeps a sharp copy of the retina, which, divided into every frame, pulls the registration back to zero. Fourier transforms are taken of wJ_k, with w a Hann window that tapers the borders to zero, so the frame edges do not register.

The Fourier shift theorem. With the discrete transform

$$\mathcal{F}\{I\}(u,v) = \sum_{x,y} I(x,y)\, e^{-2\pi i(ux/W + vy/H)}, \tag{6}$$

a translation by $\Delta = (\Delta_x, \Delta_y)$ multiplies the transform by a phase ramp:

$$I(\mathbf{x} - \Delta) \;\longleftrightarrow\; \mathcal{F}\{I\}(u,v)\, e^{-2\pi i\,\phi(\Delta)}, \qquad \phi(\Delta) = \frac{u\Delta_x}{W} + \frac{v\Delta_y}{H}. \tag{7}$$

Shift by phase correlation. Let $F_k = \mathcal{F}\{wJ_k\}$. If frame k is the reference moved by Δ_k, then $F_k \approx F_{\text{ref}}\, e^{-2\pi i \phi(\Delta_k)}$, and the normalised cross-power spectrum keeps only the phase ramp:

$$R_k(u,v) = \frac{F_k(u,v)\, F^*_{\text{ref}}(u,v)}{\big|F_k(u,v)\, F^*_{\text{ref}}(u,v)\big| + \epsilon} \approx e^{-2\pi i \phi(\Delta_k)}, \qquad r_k(\mathbf{x}) = \mathcal{F}^{-1}\{R_k\}(\mathbf{x}) \approx \delta(\mathbf{x} - \Delta_k). \tag{8}$$

The shift is the position of the peak, $\Delta_k = \arg\max_{\mathbf{x}} r_k(\mathbf{x})$, with indices above $W/2$ (or $H/2$) read as negative shifts; ϵ keeps empty frequencies from dividing by zero. The peak is refined to $1/U$ px with an upsampled DFT around it, $U = 10$ (Guizar-Sicairos, Thurman and Fienup, 2008). A simpler refinement fits a parabola through the peak and its neighbours, $\delta_x = (r_{-1} - r_{+1})/\big(2(r_{-1} - 2r_0 + r_{+1})\big)$ along x, and the same along y.

Fixations. The eye holds still (fixations) between fast jumps (saccades). On frames shrunk by 4, the shift between consecutive frames, $\Delta_{k-1\to k}$, is measured by phase correlation, and the burst is cut wherever $|\Delta_{k-1\to k}| > 30$ px. This gives fixations $\Phi_1, \ldots, \Phi_F$, runs of consecutive frames.

Within a fixation. The reference is the fixation's sharpest passed frame, $r_f = \arg\max \{s_k : k \in \Phi_f,\ \text{keep}_k\}$ (any of its frames if none passed). Every frame of the fixation is registered onto it by phase correlation, to 0.1 px. Within a fixation the motion is mostly translation; where rotation matters there too, the translation is the starting point of the same ECC refinement used between fixations, with the Euclidean model.

Between fixations. Each fixation's cleaned, aligned, passed frames are averaged into A_f. The main fixation f_0 is the one with the most passed frames. Each A_f is aligned onto A_{f_0}: first a translation by phase correlation, then a refinement by ECC with one of two motion models: Euclidean (rotation and translation, since the eye can twist during a saccade) or affine (which adds scale and shear). If the ECC search fails, the translation is kept.

The transforms. Each transform maps stabilised (reference) coordinates into the original frame:

$$T(\mathbf{x}) = \mathbf{A}\mathbf{x} + \mathbf{t}, \qquad \mathbf{A} = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix} \text{ (Euclidean)}, \qquad \mathbf{A} \in \mathbb{R}^{2\times 2} \text{ (affine)}. \tag{9}$$

A frame's full transform composes its fixation's offset with its own registration, $T_k = T_k^{\text{in}} \circ T_{f(k)}^{\text{fix}}$: from the main reference to its fixation's reference, then into the frame. The stabilised frame is

$$I'_k(\mathbf{x}) = I_k\big(T_k(\mathbf{x})\big), \tag{10}$$

by cubic interpolation, and NaN where $T_k(\mathbf{x})$ falls outside the frame. The rotation is read back as $\theta_k = \text{atan2}(A_{21}, A_{11})$.

ECC. The enhanced correlation coefficient (Evangelidis and Psarakis, 2008) between the main fixation's average and a warped fixation average is

$$\rho(\mathbf{p}) = \frac{\hat{\mathbf{a}}_{f_0}^{\top}\, \hat{\mathbf{a}}_f(\mathbf{p})}{\|\hat{\mathbf{a}}_{f_0}\|\ \|\hat{\mathbf{a}}_f(\mathbf{p})\|}, \tag{11}$$

where $\hat{\mathbf{a}}$ are the images as zero-mean vectors over the pixels and $\mathbf{p}$ are the transform's parameters (θ and $\mathbf{t}$, or the six affine entries). It is maximised iteratively from the phase-correlation start (up to 200 iterations, or until $\mathbf{p}$ changes by less than 10^{-6}), after a light Gaussian smoothing. Because it is normalised, it ignores differences of brightness and contrast between the fixation averages.

Rotation in the Fourier domain. An alternative start for the rotation needs no search. The magnitude $|\mathcal{F}\{I\}|$ ignores translation, and rotating the image rotates it by the same angle. Resampled in polar coordinates, the rotation becomes a shift along the angle axis, which phase correlation measures (Reddy and Chatterji, 1996). Derotate, then phase-correlate for the translation.

Averages. The stabilised burst keeps every frame. Averages use passed frames only, each pixel over the frames that cover it. For a set $\mathcal{K}$ of passed frames (all of them, one fixation, or one cluster below),

$$\bar{I}_{\mathcal{K}}(\mathbf{x}) = \frac{\sum_{k\in\mathcal{K}} m_k(\mathbf{x})\, I'_k(\mathbf{x})}{\sum_{k\in\mathcal{K}} m_k(\mathbf{x})}, \qquad m_k(\mathbf{x}) = \big[\, I'_k(\mathbf{x}) \text{ is finite} \,\big], \tag{12}$$

and NaN where no frame of the set covers the pixel. The noise falls as $1/\sqrt{n(\mathbf{x})}$, with $n(\mathbf{x}) = \sum_{k\in\mathcal{K}} m_k(\mathbf{x})$.

3 Background clusters and optical density

Optical density. Vessels absorb light, so they are dark in $\bar{I}$. By the Beer–Lambert law the light transmitted through a path of length ℓ in a medium of attenuation μ is $I = I_0\, e^{-\mu\ell}$. With $L = \ln \bar{I}$ and a background B in log units,

$$\mathrm{OD}(\mathbf{x}) = B(\mathbf{x}) - L(\mathbf{x}) = \ln \frac{e^{B(\mathbf{x})}}{\bar{I}(\mathbf{x})} = \mu\, \ell(\mathbf{x}). \tag{13}$$

- Vessels become positive and the background about zero, and uneven illumination cancels.
- Across a round vessel of radius r, the path length is the chord, so the profile is $\mathrm{OD}(u) = 2\mu\sqrt{r^2 - u^2}$ at distance u from the centreline, before the optics' blur.
- This takes the camera's dark level as 0. If the camera adds a black offset d, subtract it from $\bar{I}$ first.

Background fit. Pixels with $0 < \bar{I} < 4095$ are good; the others take the log of the nearest good pixel and stay invalid.

1. **First guess.** B_0 is a grey closing of L (a maximum filter, then a minimum filter, over a disc of radius 60 px, on a grid 4 times coarser, then smoothed). It fills dark structures up to about 120 px wide, so the vessels vanish and the background around them remains. With $\mathrm{OD}_0 = B_0 - L$, the first background pixels are those with $G_1 * \mathrm{OD}_0 < \mathrm{med} + 2\,\hat{\sigma}$, where $\hat{\sigma} = 1.4826\,\mathrm{MAD}$ is a robust standard deviation.
2. **Refit, twice.**
   (a) Contrast-to-noise at sizes $s \in \{1, 2, 4, \ldots, 32\}$ px: the centre minus a surround twice as wide, over its local noise on background pixels, and the best size: $Z = \max_s \big(G_s * \mathrm{OD} - G_{2s} * \mathrm{OD}\big) / \mathrm{rms}_{\mathrm{bg}}$.
   (b) Vessel mask: $M = [Z > 2] \vee [\, G_1 * \mathrm{OD} > 3\,\sigma_n]$, dilated by 2 px so vessel edges do not leak into the background.
   (c) Refit B on the pixels outside M by normalised convolution over four octaves ($\sigma = 8 \cdot 2^j$ px with weights 0.03^j, $j = 0, \ldots, 3$), so wide vessels are bridged from the background around them.
   (d) $\mathrm{OD} = B - L$.

Why cluster. The light source is fixed to the camera, not to the eye. When the eye moves between fixations, the retina under a given stabilised pixel is lit differently: glare, vignetting and the illumination gradient move across it. An intensity average over fixations mixes these backgrounds, which no single smooth fit describes. Optical density removes the illumination, so it is computed for each group of fixations that share a background, and only then combined.

Clustering. *(General form; match the distance and the threshold to the clustering code.)* For each fixation f, take the average of its passed frames, $\bar{I}_{\Phi_f}$, and its background fit B_f. The distance between two fixations' backgrounds is measured over the pixels both cover, with a global offset removed, since a change of exposure is a constant in log units and cancels in the optical density:

$$D(f,g) = \mathrm{rms}_{\mathbf{x}}\Big[\, B_f(\mathbf{x}) - B_g(\mathbf{x}) - \mathrm{med}_{\mathbf{x}}\big(B_f - B_g\big)\Big]. \tag{14}$$

Agglomerative clustering with complete linkage, cut at a threshold τ, groups the fixations into clusters $\mathcal{C}_1, \mathcal{C}_2, \ldots$, within which every pair of backgrounds differs by less than τ.

Per cluster, then combined. Each cluster q gives the average of its passed frames, $\bar{I}_q$, its own background fit B_q, its optical density and that density's local noise:

$$\mathrm{OD}_q = B_q - \ln \bar{I}_q, \qquad \sigma_q(\mathbf{x}) = \text{local rms of } \mathrm{OD}_q \text{ on background pixels.} \tag{15}$$

Pixels saturated by glare in a cluster's frames are invalid in that cluster. The clusters combine by inverse variance, which also gives the noise of the result:

$$\mathrm{OD}(\mathbf{x}) = \frac{\sum_q w_q(\mathbf{x})\,\mathrm{OD}_q(\mathbf{x})}{\sum_q w_q(\mathbf{x})}, \qquad w_q(\mathbf{x}) = \frac{1}{\sigma_q^2(\mathbf{x})}, \qquad \sigma_n(\mathbf{x}) = \Big(\sum_q w_q(\mathbf{x})\Big)^{-1/2}, \tag{16}$$

with $w_q = 0$ where cluster q has no valid data. Thresholds below are in units of σ_n.

4 Scale space

Smooth the optical density with Gaussians and take derivatives with Gaussian-derivative kernels:

$$G_\sigma(\mathbf{x}) = \frac{1}{2\pi\sigma^2}\, e^{-|\mathbf{x}|^2/2\sigma^2}, \qquad L_{xx} = \frac{\partial^2 G_\sigma}{\partial x^2} * \mathrm{OD}, \quad L_{yy} = \frac{\partial^2 G_\sigma}{\partial y^2} * \mathrm{OD}, \quad L_{xy} = \frac{\partial^2 G_\sigma}{\partial x\,\partial y} * \mathrm{OD}. \tag{17}$$

- **Invalid pixels.** First fill them by normalised convolution, $\widehat{\mathrm{OD}} = \big(G_\sigma * (m \cdot \mathrm{OD})\big)/(G_\sigma * m)$, with m the validity mask, then take derivatives, and treat pixels within about 3σ of invalid ones as unreliable. Otherwise every border and gap becomes a false edge.
- **Scales.** Steps of $\sqrt{2}$: $\sigma_j = \sigma_0\, 2^{j/2}$, $j = 0, \ldots, J-1$. With $\sigma_0 = 1$ px and $J = 8$ this is $1, 1.4, 2, 2.8, 4, 5.7, 8, 11.3$ px, which covers vessel widths of roughly 2 to 30 px (section 7).

5 The Hessian and its eigenvalues

At every pixel and scale,

$$\mathcal{H}_\sigma(\mathbf{x}) = \sigma^2 \begin{pmatrix} L_{xx} & L_{xy} \\ L_{xy} & L_{yy} \end{pmatrix}. \tag{18}$$

The factor σ^2 is Frangi's scale normalisation: it makes responses at different scales comparable (section 7). Its eigenvalues are

$$\lambda_\pm = \sigma^2 \left[\frac{L_{xx} + L_{yy}}{2} \pm \sqrt{\left(\frac{L_{xx} - L_{yy}}{2}\right)^2 + L_{xy}^2} \right], \tag{19}$$

relabelled λ_1, λ_2 so that $|\lambda_1| \le |\lambda_2|$. For a bright vessel on the optical density, the eigenvector of λ_1 points along the vessel and that of λ_2 across it; the vessel's direction is

$$\theta = \tfrac{1}{2}\,\mathrm{atan2}\big(2L_{xy},\; L_{xx} - L_{yy}\big). \tag{20}$$

structure (on OD)	λ_1	λ_2
background	≈ 0	≈ 0
vessel (bright line)	≈ 0	strongly negative
blob	strongly negative	strongly negative
dark line (not a vessel)	≈ 0	strongly positive

6 Frangi vesselness at one scale

Two measures (Frangi, Niessen, Vincken and Viergever, 1998):

$$R_B = \frac{\lambda_1}{\lambda_2}, \qquad S = \sqrt{\lambda_1^2 + \lambda_2^2}. \tag{21}$$

R_B is near 0 for a line and near 1 for a blob; S is near 0 on flat background. The two are combined as

$$V_\sigma(\mathbf{x}) = \begin{cases} 0 & \lambda_2 > 0, \\ \exp\!\left(-\dfrac{R_B^2}{2\beta^2}\right)\left(1 - \exp\!\left(-\dfrac{S^2}{2c^2}\right)\right) & \text{otherwise.} \end{cases} \tag{22}$$

- $\beta = 0.5$ is the usual value. The 3-D filter has a third term, for plate-like structures, with no 2-D counterpart.
- Frangi sets c to half the largest S in the image, which makes the result depend on the image's content. Here c is set in noise units instead: $c = \kappa\, S_{\mathrm{noise}}(\sigma)$, with $S_{\mathrm{noise}}(\sigma)$ the root-mean-square S of pure noise of level σ_n at scale σ (computed once by simulation), and $\kappa \approx 3$ to start. The same issue affects the frame gate (section 1).
- On raw intensity, where vessels are dark, the sign test flips: $V_\sigma = 0$ where $\lambda_2 < 0$. In scikit-image: `skimage.filters.frangi(image, sigmas=..., black_ridges=True)`.

7 Across scales, and the width

$$V(\mathbf{x}) = \max_j V_{\sigma_j}(\mathbf{x}), \qquad \sigma^*(\mathbf{x}) = \arg\max_{\sigma_j} V_{\sigma_j}(\mathbf{x}). \tag{23}$$

For a continuous σ^*, fit a parabola in $\log\sigma$ through the best scale and its two neighbours.

Why σ^2. Take a vessel whose cross-section is a Gaussian of height A and standard deviation s_0, $f(u) = A\,e^{-u^2/2s_0^2}$. Smoothing gives $A\,s_0(s_0^2+\sigma^2)^{-1/2}\,e^{-u^2/2(s_0^2+\sigma^2)}$, whose second derivative at the centre is $-A\,s_0\,(s_0^2+\sigma^2)^{-3/2}$. With the normalisation $\sigma^{2\gamma}$ the response is

$$\rho(\sigma) = A\,s_0\,\sigma^{2\gamma}\left(s_0^2+\sigma^2\right)^{-3/2}, \qquad \frac{d\rho}{d\sigma} = 0 \implies \sigma^{*2} = \frac{2\gamma}{3-2\gamma}\,s_0^2. \tag{24}$$

- Frangi's $\gamma = 1$ gives $\sigma^* = \sqrt{2}\,s_0$ and a peak height $\rho(\sigma^*) = 2A/3^{3/2} \approx 0.385\,A$ that does not depend on s_0: wide and narrow vessels of the same contrast score the same.
- Lindeberg's $\gamma = 3/4$ for ridges gives $\sigma^* = s_0$, but a peak height proportional to $s_0^{-1/2}$. Only $\gamma = 1$ makes the peak height independent of the width.

Real vessels. The chord profile of a round lumen is not Gaussian, so the relation between σ^* and the radius r must be calibrated. Matching second moments gives the rough guide $s_0^2 \approx r^2/4 + \sigma_b^2$, with σ_b the optics' blur; the numerical optimum differs from it for wide, sharp lumens:

radius r (px)	blur σ_b (px)	σ^*, numerical	σ^*, moment guide $\sqrt{2(r^2/4 + \sigma_b^2)}$
4	0	3.17	2.83
4	1.5	3.62	3.54
8	0	6.57	5.66
8	1.5	6.78	6.04
15	0	12.41	10.61
15	1.5	12.54	10.82

For $r \gg \sigma_b$, $\sigma^* \approx 0.8\,r$. In practice: tabulate $\sigma^*(r, \sigma_b)$ once on synthetic profiles, and invert it to read the radius from σ^*. Below about twice the blur every vessel has nearly the same profile shape, and the width must come from the contrast (the area under the profile) instead.

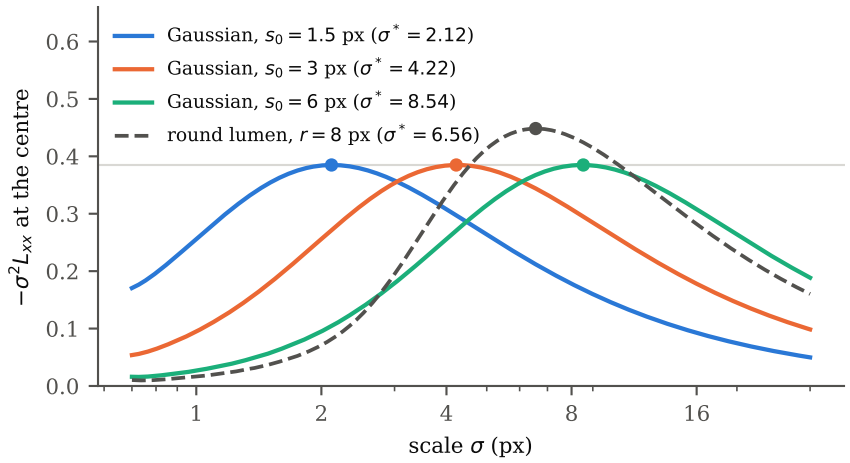


Figure 1: Scale selection with the σ^2 normalisation. Gaussian profiles of three widths peak at $\sigma^* = \sqrt{2}\,s_0$ with the same height, $2/3^{3/2}$ (grey line). A round lumen of radius 8 px (dashed, normalised to the same peak) selects $\sigma^* \approx 0.8\,r$.

8 From vesselness to a graph

This is the Hessian baseline.

1. **Centrelines.** Keep the ridge of V: pixels at least as high as their two neighbours across the vessel, $V(\mathbf{x}) \geq V(\mathbf{x} \pm \mathbf{n})$, where $\mathbf{n}$ is the unit eigenvector of λ_2 (non-maximum suppression, with sub-pixel interpolation).
2. **Hysteresis.** A ridge pixel above t_{high} starts a centreline; neighbours above t_{low} continue it.
3. **Graph.** Link ridge pixels into curves. Nodes are ends (one neighbour) and branch points (three or more). Prune spurs shorter than about one vessel width.
4. **Widths** along each centreline from σ^*, through the calibration of section 7.

9 Where the Hessian chain fails

$\mathcal{H}_\sigma$ is a symmetric 2×2 matrix: it has one pair of axes, so it can describe *one* orientation per pixel. Where two vessels meet, that is not enough. Figure 2 applies the chain above (scales 1 to

11.3 px, $\beta = 0.5$, c in noise units) to a synthetic optical-density image with three junctions.

- **Two equal vessels crossing at about 90°.** At the centre both eigenvalues are large and negative, so the pixel looks like a blob: $V = 0.20$ there, against 1.0 along the arms. The skeleton of $V > 0.5$ becomes a tangle (53 branch pixels and 6 small loops within 14 px) instead of one four-way node.
- **A fork.** The two branches stay merged in V for some distance past the split, so the skeleton's branch point lands 14 px downstream of the true fork (the trunk is 8 px wide).
- **A thin vessel crossing a wide one.** Here the topology survives (four arms leave the crossing), but inside the wide vessel V and σ^* belong to the wide vessel: the thin vessel's width and contrast are not measured where it crosses.

Oriented filters (orientation scores: one response per orientation at every pixel and scale) can hold two orientations at the same pixel, which is what junctions need. In this chain the Hessian serves as the first layer for plain vessel stretches, scale and width, and as the baseline; oriented cells take over where vessels meet.

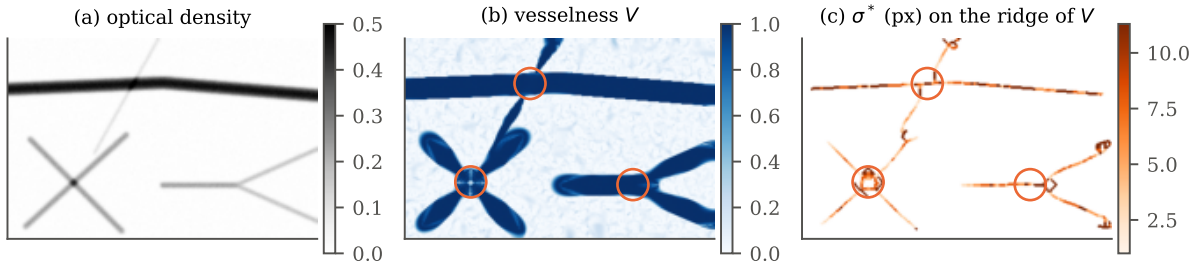


Figure 2: Frangi vesselness on a synthetic optical-density image (round lumens, blur 1.2 px, noise 0.01): a thin vessel crossing a wide one (top), two equal vessels crossing (bottom left), and a fork (bottom right). (a) The optical density. (b) Vesselness V; it falls to 0.20 at the centre of the equal crossing. (c) The selected scale σ^* on the skeleton of $V > 0.5$; wide vessels select larger scales. Circles mark the junctions.

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